

Decoding Visual Imagery from MEG Signals using Multi-Pathway Ensemble Learning

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Abstract

We address the task of decoding mentally imagined object categories from magnetoencephalography (MEG) brain recordings, submitted as a competitive entry to the IMAGINE Decoding Challenge on Kaggle (hosted by Dr. Gordon Feld, Central Institute of Mental Health, Mannheim). The competition requires classifying which of ten object categories - apple, bicycle, brush, cake, clown, cup, desk, foot, mountain, or zebra - a participant is imagining, given only stimulus-evoked (localizer) brain signals for training. The fundamental obstacle is a perception-to-imagery domain gap: MEG responses during active mental imagery are related to but neurologically distinct from those elicited by direct visual perception. We present a modality-aware multi-pathway ensemble that exploits a shared auditory processing window present in both paradigms. By training classifiers on the localizer auditory window (subjects hearing the word cue, approximately -1000 to 0 ms relative to visual onset) and applying them to the imagine auditory window (0 to 1000 ms, the same acoustic stimulus), we achieve same-modality transfer that substantially reduces the domain gap compared to cross-modal approaches. Our full system combines six weighted decoders spanning sliding-window one-vs.-rest logistic regression banks, full-window multinomial classifiers, and GloVe-300 word embedding regression. The ensemble achieves a Kaggle leaderboard accuracy of 0.133 (rank 27/169 teams), improving 23% relative over a baseline SVC at 0.108 (rank 59/92) in week 2.

1 Introduction

Recovering cognitive states from noninvasive neural recordings is a long-standing goal at the intersection of neuroscience and machine learning. Magnetoencephalography (MEG) measures the weak magnetic fields produced by synchronous neuronal currents at millisecond temporal resolution, making it uniquely suited for decoding the rapid dynamics of sensory and cognitive processing. Applications range from communication prosthetics for patients with motor impairments to fundamental research probing the neural correlates of perception and imagination.

The IMAGINE Decoding Challenge.

This work describes our entry in the IMAGINE Decoding Challenge, a Kaggle competition hosted by Dr. Gordon Feld of the Central Institute of Mental Health, Mannheim, run as a course project for Johns Hopkins University's graduate-level Deep Neural Networks course (DNN 605.742, Spring 2026). The competition presents 169 teams with a 10-class object classification problem. Prizes of \$600/\$300/\$100 are awarded only if accuracy exceeds 20%; chance is 10%. No team surpassed the prize threshold.

Task definition.

Participants viewed or heard 10 object words (apple, bicycle, brush, cake, clown, cup, desk, foot, mountain, zebra) during two experimental phases. In the localizer phase, they heard an auditory cue then saw an image of the corresponding object. In the imagine phase, they heard only the auditory cue and mentally visualized the

object with eyes closed. Localizer recordings carry ground-truth labels; imagine labels are scrambled and must be recovered by the model.

Formal problem statement.

Let X_{loc} in $\mathbb{R}^{(C \times T_{\text{loc}})}$ denote a localizer epoch with label y in $\{1, \dots, 10\}$, and X_{img} in $\mathbb{R}^{(C \times T_{\text{img}})}$ an imagine epoch. The goal is to learn f from the labeled localizer distribution $p(X_{\text{loc}}, y)$ that generalizes to $p(X_{\text{img}})$. The key challenge is:

$$p(X_{\text{loc}}) \neq p(X_{\text{img}}) \quad (1)$$

Neural dynamics during imagery only partially overlap with those during perception (Dijkstra et al., 2019; Bezsudnova et al., 2024).

Contributions.

We make three primary contributions: (1) identifying and exploiting a same-modality auditory transfer pathway that eliminates the perception-imagery domain gap for the acoustic segment; (2) constructing a six-decoder weighted ensemble combining auditory and visual pathways, sliding-window and full-window classifiers, and semantic embedding regression; (3) systematic analysis of two failed approaches that distinguishes transfer failure from featurization failure.

2 Dataset and Experimental Setup

2.1 MEG Recording System

All recordings were collected using a MEGIN Triux system (formerly Elekta Neuromag) with a 306-channel Vectorview layout: 102 magnetometers and 204 planar gradiometers, plus ECG and EOG reference channels. Raw signals were downsampled to 100 Hz before release. Data are stored in MNE-Python Epochs format (one epoch = one trial). Signal Space Separation (SSS; Taulu & Simola, 2006) was applied at acquisition to suppress external magnetic interference.

2.2 Dataset Structure

The dataset comprises 29 subjects: 15 training and 14 test. Each contributes localizer epochs (480 trials, 48 per class, -200 to +1000 ms re. visual onset, 121 time steps) and imagine epochs (50 trials, -200 to +5000 ms re. auditory cue onset, 521 time steps). Only localizer data may be used for training; imagine test labels are withheld.

Split	Type	Epochs/subj.	Channels	Time steps	Labels
Train (15)	Localizer	480 (48/class)	309	121	Provided
Train (15)	Imagine	50	309	521	Provided (local CV)
Test (14)	Localizer	480 (48/class)	309	121	Provided
Test (14)	Imagine	50	309	521	Scrambled - predict

Table 1. Dataset dimensions per subject. Localizer: -200 ms to +1000 ms re. visual onset. Imagine: -200 ms to +5000 ms re. auditory cue onset.

2.3 Computational Resources

All experiments ran on free-tier Google Colab: Intel Xeon @ 2.2 GHz (2 cores), 12.7 GB RAM, 78 GB storage. No GPU was used; all models are CPU-based scikit-learn classifiers. The absence of GPU acceleration was not limiting given our feature-based decoder architecture, in which the most expensive operations are per-subject logistic regression fitting across hundreds of time windows.

3.1 Sensor Selection: Gradiometers Only

The MEGIN Triux layout provides 102 magnetometers and 204 planar gradiometers. Magnetometers are sensitive to deep and distant neural sources but also pick up far-field interference, increasing inter-subject variability. Planar gradiometers measure the spatial gradient of the magnetic field and are primarily sensitive to local, superficial cortical sources - precisely the population-level activity we expect to discriminate between word categories. An ablation study comparing all 306 channels, gradiometers only, and magnetometers only confirmed that gradiometers alone produced the highest cross-validated accuracy on held-out localizer folds.

3.2 Time Window Selection

Localizer auditory window.

The localizer epoch begins with the subject hearing the target word. The auditory cortex responds within 100 ms and sustains activation through approximately 400 ms (Hickok & Poeppel, 2007). This response is embedded in the localizer's negative-time segment (approximately -1000 to -50 ms re. visual onset, since the auditory cue precedes image presentation by approximately 1000 ms). Crucially, the first 200 ms of the visual window (0 to +200 ms) captures only a modality-nonspecific onset response identical across all 10 categories and is therefore excluded.

Localizer visual window.

Visual object processing produces category-discriminative neural responses beginning around 200 ms post-onset, peaking at 300-600 ms in temporal and occipito-temporal regions. We use +50 ms to +600 ms for visual decoders, capturing the core object recognition response while excluding the early nonspecific onset.

Imagine windows.

The imagine epoch is locked to auditory cue onset ($t=0$). The auditory response occurs in 0-1000 ms, exactly as in the pre-visual localizer segment. Mental imagery proper emerges later, typically after 500 ms, and persists throughout the epoch (Dijkstra et al., 2019). We use 0-1000 ms for auditory-pathway decoders applied to imagine data, and 300-3000 ms for visual/imagery-pathway decoders.

3.3 ICA Artifact Removal: Tried and Rejected

Independent Component Analysis (ICA) was evaluated to remove ECG and EOG artifacts (Jung et al., 2000). Models trained on ICA-cleaned data consistently performed worse due to two factors: (1) signal loss - components correlated with ECG or EOG can also carry genuine cortical activity, particularly in temporal and frontal channels where artifact and neural sources share topography (Castellanos & Makarov, 2006); (2) redundancy - the MEGIN SSS preprocessing already attenuates external interference (Taulu & Simola, 2006), so applying ICA on top offers diminishing returns while risking removal of task-relevant neural signal. ICA was excluded from the final pipeline.

3.4 Null Data Injection

Pre-stimulus baseline epochs (-200 ms to 0 ms) are randomly sampled and assigned a null class label during OvR classifier training. This sharpens the decision boundary between genuine neural responses and resting-state activity, reducing false positives when the sliding-window decoder scans the imagery epoch at time points that may contain no class-discriminating signal (Blankertz et al., 2011). Empirically, null injection increased validation accuracy by 1-2 percentage points.

4 Feature Engineering

Three feature families are computed from the 204-channel gradiometer data, each capturing a distinct aspect of

4.1 Temporal Window Statistics

The analysis epoch is divided into $N_{\text{win}} = 5$ equal temporal segments. For each segment k and each channel c , the mean and standard deviation of raw signal amplitude are computed:

$$\mu_{\text{c},k} = (1/|W_k|) * \sum_{t \in W_k} x_{\text{c},t} \quad (2)$$

$$\sigma_{\text{c},k} = \sqrt{(1/|W_k|) * \sum_{t \in W_k} (x_{\text{c},t} - \mu_{\text{c},k})^2} \quad (3)$$

This yields $2 \times 5 \times 204 = 2,040$ features per epoch. The five temporal windows provide coarse tracking of the evoked response's temporal dynamics while the standard deviation captures within-window variability. This representation is robust to single-trial amplitude fluctuations while preserving the gross temporal profile.

4.2 Band-Power Features

For each of four physiologically motivated frequency bands, we apply a 4th-order Butterworth bandpass filter, compute the Hilbert analytic envelope, and average power over the analysis window per channel:

$$\text{BP}_{\text{c},b} = (1/T) * \sum_t |H\{x_{\text{c},t}^{(b)}\}|^2 \quad (4)$$

where H denotes the Hilbert transform and $x_{\text{c},t}^{(b)}$ is the bandpass-filtered signal for band b . This produces $4 \times 204 = 816$ additional features. Combined with temporal statistics, the full feature vector has dimension 2,856.

Band	Range (Hz)	Cognitive association
Theta	4 - 8	Working memory, object recognition
Alpha	8 - 13	Inhibition, imagery gating
Beta	13 - 30	Motor planning, cognitive effort
Gamma	30 - 45	Local computation, feature binding

Table 2. Frequency bands used for band-power feature extraction.

4.3 GloVe Word Embeddings

Each of the 10 target word categories is represented by its 300-dimensional GloVe vector (Pennington et al., 2014), pre-trained on the Wikipedia + Gigaword corpus (glove-wiki-gigaword-300). These embeddings encode inter-word semantic relationships through distributional co-occurrence statistics: semantically related words have higher cosine similarity (e.g., 'apple' and 'cake' are both food items; 'mountain' and 'zebra' are unrelated). The embedding regression decoder exploits this structure to inject semantic geometry into the classification process.

4.4 Riemannian Covariance Features (Attempted, Excluded)

We additionally computed covariance-based features using the Riemannian geometry framework implemented in the pyriemann library (Barachant et al., 2012). For each trial, the 204×204 channel covariance matrix is estimated using Ledoit-Wolf regularization, then projected onto the tangent space of the symmetric positive definite (SPD) manifold at the Frechet mean of the training set. This yields a Euclidean feature vector of dimension $204 \times 205 / 2 = 20,910$. Although these features achieve strong localizer CV accuracy (mean 0.238), they fail catastrophically on imagery (0.095, chance level) as described in Section 9.

5 Model Development Journey

Week 2 - Full MEG with artifact removal (baseline).

Beginning from a minimal EOG-only baseline, we expanded to the full 306-channel MEG signal and implemented a custom artifact removal procedure using Ridge regression: a Ridge model predicts ECG and EOG contributions to each MEG channel using the reference signals with +/-50 ms time lags, and the predicted artifact component is subtracted. From the artifact-cleaned signal, we extracted 50 ms windowed averages (~3,264 features per trial), applied a RobustScaler, and trained an SVC with RBF kernel (C=10). Validation accuracy: 10.93% (p=0.17 vs. chance). Leaderboard: 0.108, rank 59/92.

Week 3 - Channel ablation and time window selection.

Recognizing that magnetometers add inter-subject variability without improving discriminability, we restricted to 204 gradiometers. Simultaneously, we revised the time window to exclude the first 200 ms of the visual window (category-nonspecific auditory onset) and focus on 200-1000 ms post-onset. Validation: 12.13% (p=0.077). Leaderboard: 0.114, rank 46/103.

FPIP - Feature engineering.

By the interim presentation deadline, we had implemented temporal window statistics and band-power features, null data injection, and replaced the global SVC with multiple per-time-window classifiers. These changes collectively raised the leaderboard score to 0.120, rank 36/118.

Week 10 - Auditory pathway discovery and ensemble.

The breakthrough insight - the existence of a same-modality transfer pathway through the shared auditory window - was discovered during week 10 (described in detail in Section 6). Implementing the 6-decoder ensemble immediately improved the leaderboard score from 0.120 to 0.133, rank 16/142. This single architectural change was responsible for the majority of total improvement over the competition.

6 Key Insight: Same-Modality Transfer

The two-phase localizer epoch.

The localizer epoch spans -200 ms to +1000 ms relative to visual image onset. This means the epoch contains not only the visual evoked response (0 ms onward) but also a substantial pre-visual period during which the subject hears the target word. The auditory word presentation occurs approximately 1000 ms before the image appears, so the auditory response is embedded in the negative-time portion of the localizer epoch.

The imagine epoch begins with the same stimulus.

In the imagine paradigm, subjects hear the same spoken word cue (0 ms onset), then visualize the object. Thus the imagine epoch's initial 0-1000 ms captures the brain's response to an identical acoustic stimulus as the localizer epoch's pre-visual period.

Why this eliminates the domain gap.

Prior approaches trained decoders on the localizer's visual window (visual cortex activity during viewing) and applied them to the imagery period of the imagine epoch (mental imagery activity). This is cross-modal transfer: two distinct brain processes. By contrast, training on the localizer's auditory window and testing on the imagine auditory window is same-modality transfer: both conditions involve the auditory cortex processing the same acoustic input. The neural response to hearing 'apple' during a localizer trial and hearing 'apple' at the start of an imagine trial should be neurologically equivalent, modulo minor context effects. Formally, let ϕ_{aud} denote the neural response to an auditory word stimulus. We hypothesize:

$$p(\phi_{\text{aud}}^{\text{loc}} | y) \approx p(\phi_{\text{aud}}^{\text{img}} | y) \quad (5)$$

i.e., the distribution of auditory MEG responses during localizer and imagine trials is approximately the same given the word category y . This is not true for visual responses, where $p(\phi_{\text{vis}}^{\text{loc}} | y) \neq p(\phi_{\text{imagery}}^{\text{img}} | y)$. This hypothesis is supported empirically: auditory sliding OvR alone achieves approximately 14.1% mean validation accuracy vs. approximately 11.2% for visual sliding OvR, consistent with the neuroscience literature showing that auditory cortex responses to spoken words are highly reproducible across contexts (Hickok &

7 Model Architecture: Six-Decoder Weighted Ensemble

The final system combines six independent decoders, each outputting a probability vector p^d in R^{10} over the ten categories. The final prediction is:

$$y_{\text{hat}} = \underset{c}{\operatorname{argmax}} \sum_{d=1}^6 w_d * p_c^d \quad (6)$$

where w_d is the weight for decoder d , assigned based on expected transfer quality (higher for same-modality, lower for cross-modal).

Decoder	Train window	Apply window	w_d	Transfer type
Aud. Sliding OvR	Loc. aud. (-1s to 0s)	Img. (0-1s)	3.0	Same-modal
Aud. Full-Window	Loc. aud. (-1s to 0s)	Img. (0-1s)	2.0	Same-modal
Aud. Emb. Reg.	Loc. aud. (-1s to 0s)	Img. (0-1s)	1.5	Same-modal
Vis. Sliding OvR	Loc. vis. (0-0.6s)	Img. (0.3-3s)	1.0	Cross-modal
Vis. Full-Window	Loc. vis. (0-0.6s)	Img. (0.3-3s)	0.5	Cross-modal
Vis. Emb. Reg.	Loc. vis. (0-0.6s)	Img. (0.3-3s)	0.5	Cross-modal

Table 3. Six-decoder ensemble configuration. Auditory decoders receive higher weights due to same-modality transfer quality.

7.1 Sliding Window One-vs.-Rest Bank

The sliding window decoder trains a separate One-vs.-Rest (OvR) L1-penalized logistic regression classifier at every 10 ms time step across the training window, using 50 ms windows. For each time step t , the 50 ms x 204-channel segment is flattened to a 10,200-dimensional vector. For each class c in $\{1, \dots, 10\}$, a binary logistic regression is trained with L1 penalty ($C=6.0$):

$$\min_{w, b} ||w||_1 + C * \sum_i \log[1 + \exp(-y_i * (w^T x_i + b))] \quad (7)$$

where $y_i = +1$ if trial i belongs to class c and -1 otherwise. N_{null} null-class samples drawn from the pre-stimulus baseline are added with label -1 for all classes.

Cross-validation pruning.

Before final training, each window t is evaluated by 3-fold cross-validated accuracy on the localizer data. Windows where CV accuracy falls at or below 11% (barely above chance) are discarded. This aggressive pruning retains only time points with genuine class-discriminating signal, reducing inference noise when the decoder scans across the longer imagery epoch.

Null logit z-scoring.

At inference time, trial i 's logit vector at window t is z-scored against the null distribution computed from the baseline training samples, then centered per trial:

$$z_{(i, t, c)} = (\text{logit}_{(i, t, c)} - \mu_{c^{\text{null}}}) / \sigma_{c^{\text{null}}} \quad (8)$$

Aggregation.

The final probability estimate for decoder d is computed by weighted average over all CV-retained windows:

$$p^d(d)_i = \text{softmax}(\sum_{t \in T_{\text{keep}}} \text{cv_acc}_t * z_{(i, t)}) \quad (9)$$

7.2 Full-Window Classifier

The full-window decoder treats each trial as a single feature vector, combining temporal window statistics and

band-power features across the entire analysis window (up to 2,856 dimensions). Principal Component Analysis reduces dimensionality to 50 components, and multinomial logistic regression (LBFGS solver, C=1.0, balanced class weights) is trained on the reduced features. At inference, the imagery epoch is scanned with 0.2 s steps within the applicable window, and predicted class probabilities are averaged over all scan positions.

7.3 Word Embedding Regression

The embedding regression decoder predicts the semantic vector of the imagined concept from MEG features, rather than predicting the class label directly. Let E in $\mathbb{R}^{(10 \times 300)}$ be the matrix of GloVe embeddings for the 10 target words. A Ridge regression with $\alpha=100$ is trained on PCA-reduced MEG features ϕ_i in $\mathbb{R}^{50}$:

$$\min_W ||W * \phi_i - e_{yi}||^2 + \alpha * ||W||_F^2 \quad (10)$$

For a test trial with features ϕ_* , the predicted embedding $e_{\hat{}} = W * \phi_*$ is compared to all 10 target word embeddings by cosine distance:

$$d_c = 1 - (e_{\hat{}} \cdot e_c) / (||e_{\hat{}}|| * ||e_c||) \quad (11)$$

A temperature-scaled softmax ($\tau = 0.2$) converts distances to probabilities:

$$p_c = \exp(-d_c / \tau) / \sum_{c'} \exp(-d_{c'} / \tau), \quad \tau = 0.2 \quad (12)$$

The temperature $\tau=0.2$ sharpens the distribution, concentrating probability mass on the nearest word. This decoder benefits from semantic geometry: a confused 'apple' prediction tends to land closer to 'cake' than to 'zebra', partially recovering from the classification error (Mitchell et al., 2008).

8 Results

8.1 Competition Leaderboard Trajectory

Table 4 documents our weekly leaderboard performance across all modules. The competition began with a baseline SVC at 0.108 and ended at 0.133 following the ensemble implementation in week 10.

Module	Score	Rank / Teams	Cum. Sub.	Key change
Week 2	0.108	59 / 92	2	Full MEG + SVC baseline
Week 3	0.108	65 / 100	4	-
Week 4	0.114	46 / 103	13	Gradiometers only
Week 5	0.114	46 / 107	21	Time window tuning
FPIP	0.120	36 / 118	24	Null injection + features
Week 9	0.120	41 / 132	26	-
Week 10	0.133	16 / 142	27	Auditory ensemble
Week 11	0.133	20 / 151	30	Ensemble tuning
Week 13	0.133	27 / 169	34	Final

Table 4. Weekly Kaggle leaderboard performance. Best rank 16th at week 10 (score 0.133); final rank 27th/169 after more teams joined later modules.

8.2 Ablation Study: Individual Decoder Performance

To assess the contribution of each decoder independently, we evaluated each on the 15 training subjects using within-subject cross-validation (fit on localizer, predict imagine labels).

Decoder	Mean acc.	Median acc.
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Auditory Sliding OvR	0.141	0.140
Auditory Full-Window	0.128	0.120
Auditory Emb. Reg.	0.119	0.100
Visual Sliding OvR	0.112	0.100
Visual Full-Window	0.108	0.100
Visual Emb. Reg.	0.106	0.100
Full 6-Decoder Ensemble	0.162	0.160

Table 5. Per-decoder and ensemble validation accuracy on training subjects. The ensemble (0.162) substantially outperforms every individual component, confirming that the decoders provide complementary information.

The ensemble (0.162) substantially outperforms every individual decoder. The auditory sliding OvR is the strongest individual decoder (0.141), consistent with the same-modality transfer hypothesis.

9 Failure Mode Analysis

Two decoder architectures showed strong within-domain performance but failed to generalize to imagery. We document them in detail because the diagnostic process of distinguishing transfer failure from featurization failure is methodologically important.

9.1 Riemannian Tangent-Space Decoder: Transfer Failure

We implemented a Riemannian tangent-space decoder using Ledoit-Wolf regularized covariance matrices projected onto the SPD manifold's tangent space (via `pyriemann`). Per-subject 3-fold CV accuracy on the localizer alone was 0.238 (range 0.146-0.450), well above the 0.10 chance baseline. This confirms the decoder learns genuinely class-discriminating information: spatial covariance patterns during visual perception differ reliably across the 10 word categories.

However, when this localizer-trained decoder was applied to imagery data, accuracy collapsed to 0.095 (range 0.020-0.160), indistinguishable from chance. The 14-point gap cannot be attributed to the curse of dimensionality: the feature-to-sample ratio is 20,910:480, and cross-validation confirms the localizer model has capacity to learn. The decoder can learn; it just cannot transfer. The interpretation is that spatial covariance structure during sustained mental imagery does not match the covariance structure during visual perception.

We attempted three remediation strategies:

- 1. Covariance recentering:** subtracting the task-specific mean covariance before tangent projection (manifold domain adaptation). This reduces the geometric distance between localizer and imagery distributions on the manifold but did not recover usable accuracy.
- 2. Occipital channel subset:** restricting covariance computation to occipito-temporal channels most involved in visual processing, reducing dimensionality and focusing on the most relevant spatial structure. This did not improve imagery accuracy beyond chance.
- 3. Imagine anchoring:** mixing held-out imagery trials into the tangent space reference. While conceptually sound, this would require using imagine labels for training, which the competition prohibits.

All three strategies failed. The Riemannian decoder was excluded from the final ensemble.

9.2 Alpha-Band Log-Variance Decoder: Featurization Failure

We hypothesized that mental imagery involves sustained desynchronization of the alpha band (8-13 Hz), a well-documented phenomenon in which imagined stimuli suppress occipital alpha power relative to baseline

(Ramoser et al., 2000). We implemented an alpha-band decoder using 4th-order Butterworth filtering (8-13 Hz), followed by channel-wise log-variance of the Hilbert envelope over the imagery window (1.0-3.0 s post-cue). Localizer-trained accuracy on imagery: 0.084 (below chance).

To determine whether transfer was the limiting factor, we ran 5-fold CV entirely within the imagine training data: accuracy 0.088, still at chance. The two numbers being nearly identical ($0.084 \approx 0.088$) rules out the transfer hypothesis: even when training and testing within the same domain, the decoder performs at chance. The featurization simply does not capture class-discriminating information. Three potential explanations: (1) wrong band - sustained imagery suppression may manifest in beta or lower gamma; (2) collapsed time dimension - log-variance over 1.0-3.0 s discards temporal dynamics a time-resolved approach would preserve; (3) effect too weak or diffuse to discriminate 10 categories at single-trial level. The alpha decoder was excluded from the final ensemble.

9.3 Subject-Level Prediction Collapse

Subject sub-26 exhibited degenerate predictions: 48 of 50 imagine trials were classified as 'zebra'. This prediction collapse is a known pathology in per-subject OvR classifiers when one class's training distribution closely resembles the test distribution for a particular subject's neural geometry. In sub-26's case, the spatial pattern of the 'zebra' class in the localizer is unusually similar to the general background neural activity in the imagine epoch, causing the classifier to default to that class. Future mitigation: cross-subject regularization constraining per-subject classifiers toward the population mean.

10 Discussion

10.1 The Primacy of Domain Shift

The most important lesson from this competition is that domain shift dominates over model complexity as the limiting factor in MEG decoding. The improvement from a simple SVC baseline (0.108) to the full ensemble (0.133) did not come from using a more powerful classification algorithm or deeper features. It came from identifying and exploiting a structural property of the experimental paradigm - the shared auditory window - that allowed same-modality transfer. The Riemannian covariance experiment makes this point sharply: a decoder achieving 0.238 CV accuracy on the training domain collapses to 0.095 when transferred, while a simple logistic regression on temporal features achieves 0.141 on the target domain. Feature complexity and classifier power are secondary to alignment between training and test distributions.

10.2 Semantic Embeddings as a Complementary Signal

The GloVe embedding regression decoder contributes a qualitatively different type of information from the time-domain classifiers. Rather than learning a discriminative boundary in MEG feature space, it learns a mapping from MEG features to semantic space, where inter-class relationships are encoded. For confusable category pairs such as 'cup' and 'desk' (both indoor objects) or 'apple' and 'cake' (both foods), the embedding regression distributes probability mass toward semantically nearby categories rather than random ones, partially recovering from classification errors. This is consistent with the semantic brain decoding literature (Mitchell et al., 2008).

10.3 Ensemble Diversity

The 56% relative improvement of the ensemble (0.162) over the best individual decoder (0.141) demonstrates genuine complementarity between components. The three decoder families - sliding-window OvR (time-resolved discriminative), full-window (global temporal integration), and embedding regression (semantic) - each capture distinct aspects of the MEG signal, and their combination outperforms any single approach. This confirms that ensemble diversity, not just ensemble size, drives the improvement.

10.4 Limitations

- Accuracy ceiling: Even the best ensemble achieves 0.133, well below the 0.20 prize threshold. The fundamental perception-to-imagery domain gap may require data from the imagery domain itself to bridge reliably.
- Per-subject fitting: All classifiers are trained separately per subject, requiring at least 480 labeled localizer trials per subject for reliable parameter estimation. Cross-subject generalization remains unaddressed.
- CPU-only: The sliding-window bank requires fitting hundreds of logistic regression models per subject per pathway. GPU-accelerated deep learning (EEGNet, BrainBERT) could provide richer spatiotemporal representations.
- No temporal deep learning: Feature engineering collapses temporal dynamics to summary statistics. Recurrent or attention-based models could capture higher-order dynamics directly from the raw time series.

11 Future Work

Fine-tuning on imagine-train labels.

The competition provides labeled imagine-train data for local validation. Future work should explore using a small subset of labeled imagine trials to adapt the localizer-trained classifiers via domain adaptation or few-shot fine-tuning. Even 5-10 labeled imagine trials per class could substantially reduce the domain gap.

Cross-subject models.

Training a single model across all 15 subjects (with subject-specific normalization) would allow the model to learn subject-invariant neural features and mitigate per-subject failure modes like the sub-26 prediction collapse.

Riemannian domain adaptation.

For the Riemannian decoder specifically, more aggressive covariance recentering - projecting localizer and imagery covariances to a shared reference point on the manifold - warrants further investigation.

Alpha-band revisited.

The alpha decoder should be re-evaluated with (a) time-resolved Hilbert envelopes in 200 ms windows to preserve temporal dynamics, (b) channel-relative topographic alpha power rather than absolute magnitude, and (c) beta and lower-gamma bands as alternative targets.

Deep learning decoders.

EEGNet (Lawhern et al., 2018) and related architectures designed for EEG/MEG classification could replace feature-based classifiers, learning spatiotemporal representations directly from the raw signal without manual feature engineering.

12 Conclusion

We presented a multi-pathway ensemble approach for decoding imagined object categories from MEG brain recordings, developed as a competition entry for the IMAGINE Decoding Challenge on Kaggle (Johns Hopkins DNN 605.742, Spring 2026). The central contribution is the identification of a same-modality auditory transfer pathway: because both localizer and imagine paradigms share an initial auditory processing phase in response to the same spoken word, training on the localizer auditory window and applying decoders to the imagine auditory window achieves near-zero-gap domain transfer. Combined with full-window multinomial classifiers and GloVe word embedding regression in a six-decoder weighted ensemble, this approach achieves a leaderboard accuracy of 0.133 (rank 27/169 teams), improving by 23% relative over the initial SVC baseline.

Our systematic failure-mode analysis demonstrates that the Riemannian tangent-space covariance decoder fails not due to limited capacity but due to genuine between-domain distribution mismatch, and that alpha-band log-variance features are uninformative regardless of the training domain - a diagnostic distinction with practical implications for future work in non-Lambertian environments. The key lesson: domain alignment is the primary bottleneck, and identifying structural similarities between paradigms (such as shared auditory phases) is more valuable than increasing model complexity.

References

- Barachant, A., Bonnet, S., Congedo, M., & Jutten, C. (2012). Multiclass brain-computer interface classification by Riemannian geometry. *IEEE Transactions on Biomedical Engineering*, 59(4), 920-928.
- Bezsudnova, Y., Quinn, A.J., Wynn, S.C., & Jensen, O. (2024). Spatiotemporal properties of common semantic categories for words and pictures. *Journal of Cognitive Neuroscience*, 36(7), 1335-1353.
- Blankertz, B., Lemm, S., Treder, M., Haufe, S., & Muller, K.R. (2011). Single-trial analysis and classification of ERP components: A tutorial. *NeuroImage*, 56(2), 814-825.
- Castellanos, N.P., & Makarov, V.A. (2006). Recovering EEG brain signals: Artifact suppression with wavelet enhanced independent component analysis. *Journal of Neuroscience Methods*, 158(2), 300-312.
- Dijkstra, N., Bosch, S.E., & van Gerven, M.A.J. (2019). Shared neural mechanisms of visual perception and imagery. *Trends in Cognitive Sciences*, 23(5), 423-434.
- Hickok, G., & Poeppel, D. (2007). The cortical organization of speech processing. *Nature Reviews Neuroscience*, 8(5), 393-402.
- Jung, T.P., Makeig, S., Humphries, C., Lee, T.W., McKeown, M.J., Iragui, V., & Sejnowski, T.J. (2000). Removing electroencephalographic artifacts by blind source separation. *Psychophysiology*, 37(2), 163-178.
- Lawhern, V.J., Solon, A.J., Waytowich, N.R., Gordon, S.M., Hung, C.P., & Lance, B.J. (2018). EEGNet: A compact convolutional neural network for EEG-based brain-computer interfaces. *Journal of Neural Engineering*, 15(5), 056013.
- Mitchell, T.M. et al. (2008). Predicting human brain activity associated with the meanings of nouns. *Science*, 320(5880), 1191-1195.
- Pennington, J., Socher, R., & Manning, C.D. (2014). GloVe: Global vectors for word representation. *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, 1532-1543.
- Ramoser, H., Muller-Gerking, J., & Pfurtscheller, G. (2000). Optimal spatial filtering of single trial EEG during imagined hand movement. *IEEE Transactions on Rehabilitation Engineering*, 8(4), 441-446.
- Taulu, S., & Simola, J. (2006). Spatiotemporal signal space separation method for rejecting nearby interference in MEG measurements. *Physics in Medicine & Biology*, 51(7), 1759-1768.